

Mechanical Testing of a Spinal Fusion System

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ABSTRACT

- Mechanical instability of the spine may be caused by several issues such as degenerative disease and trauma
- To help with the pain cause by spinal instability, surgeons fuse the vertebrae together
- This is done through the insertion of a device consisting of pedicle screws and long metallic rods to hold the vertebrae together; over time the vertebrae with fuse together greatly strengthening the spine
- Following ASTM standards, operating procedures were developed to test the mechanical properties of a new design of one of these devices using an INSTRON materials testing machine

BACKGROUND

- Spinal fusions are done for multiple reasons
 - To prevent the surrounding tissues from putting too much stress of the spinal cord
 - Treat spinal stenosis, herniated disks, injuries, infections and tumors
- The number of spinal fusions done from 1998 to 2008 increased 2.4 fold

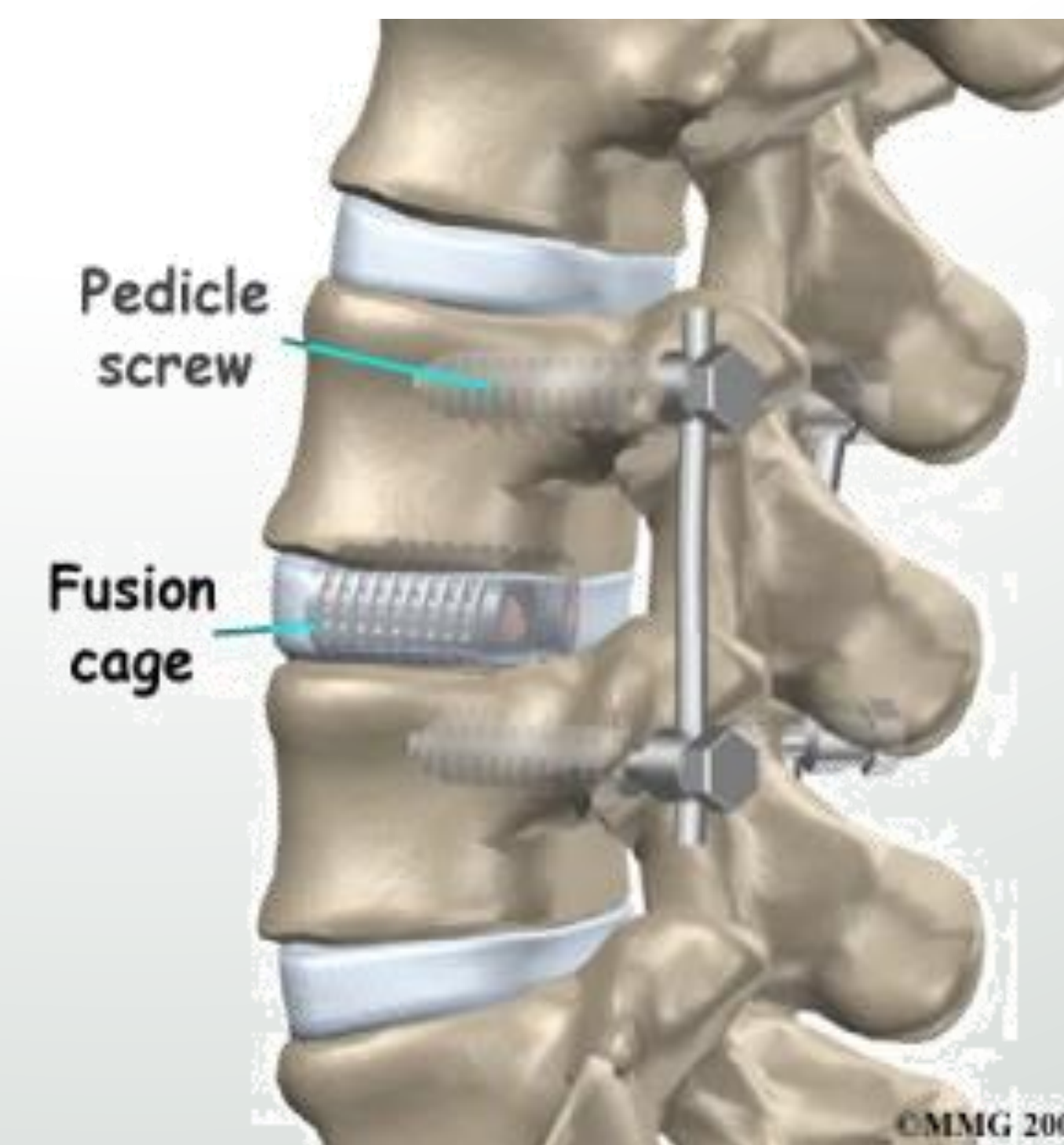
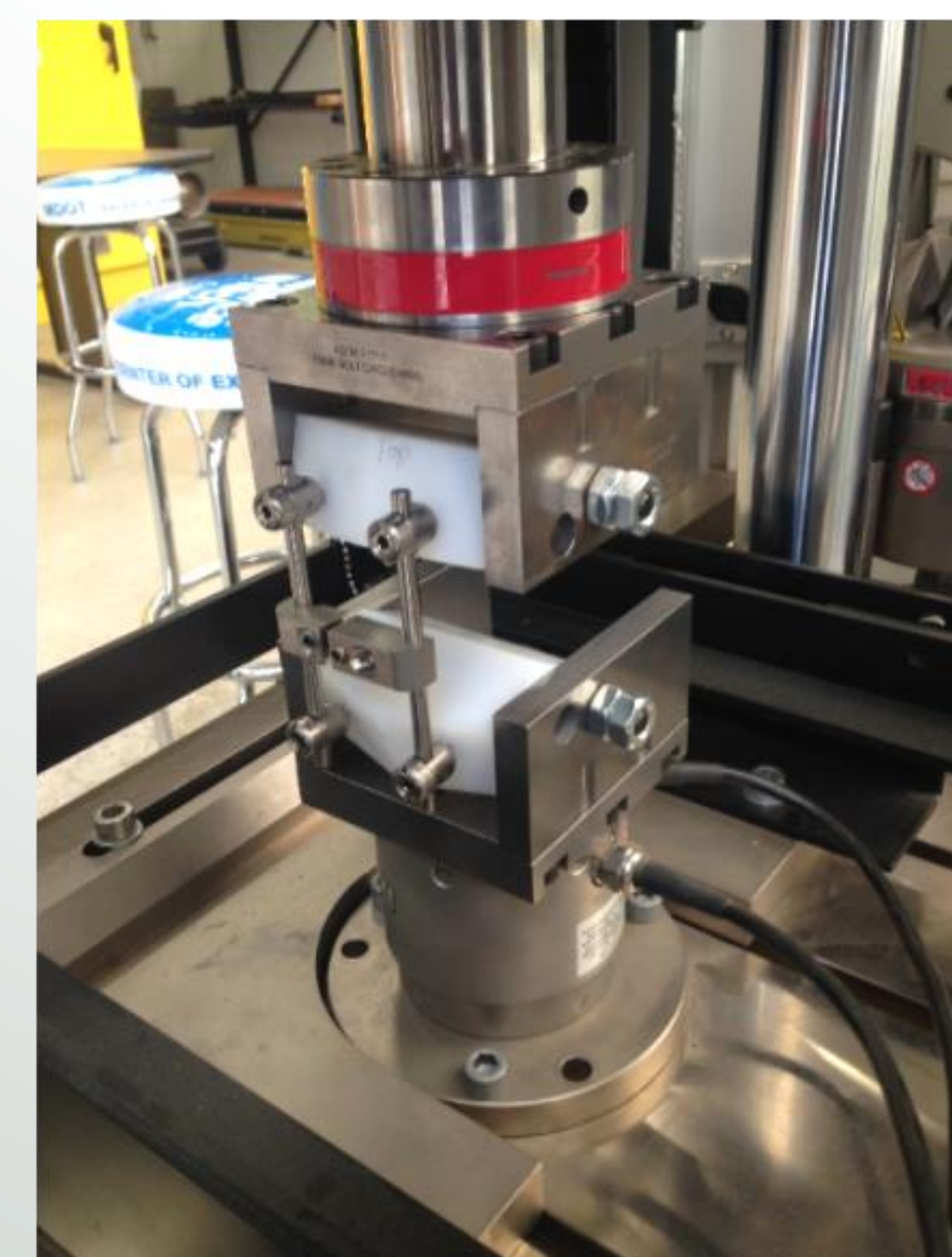
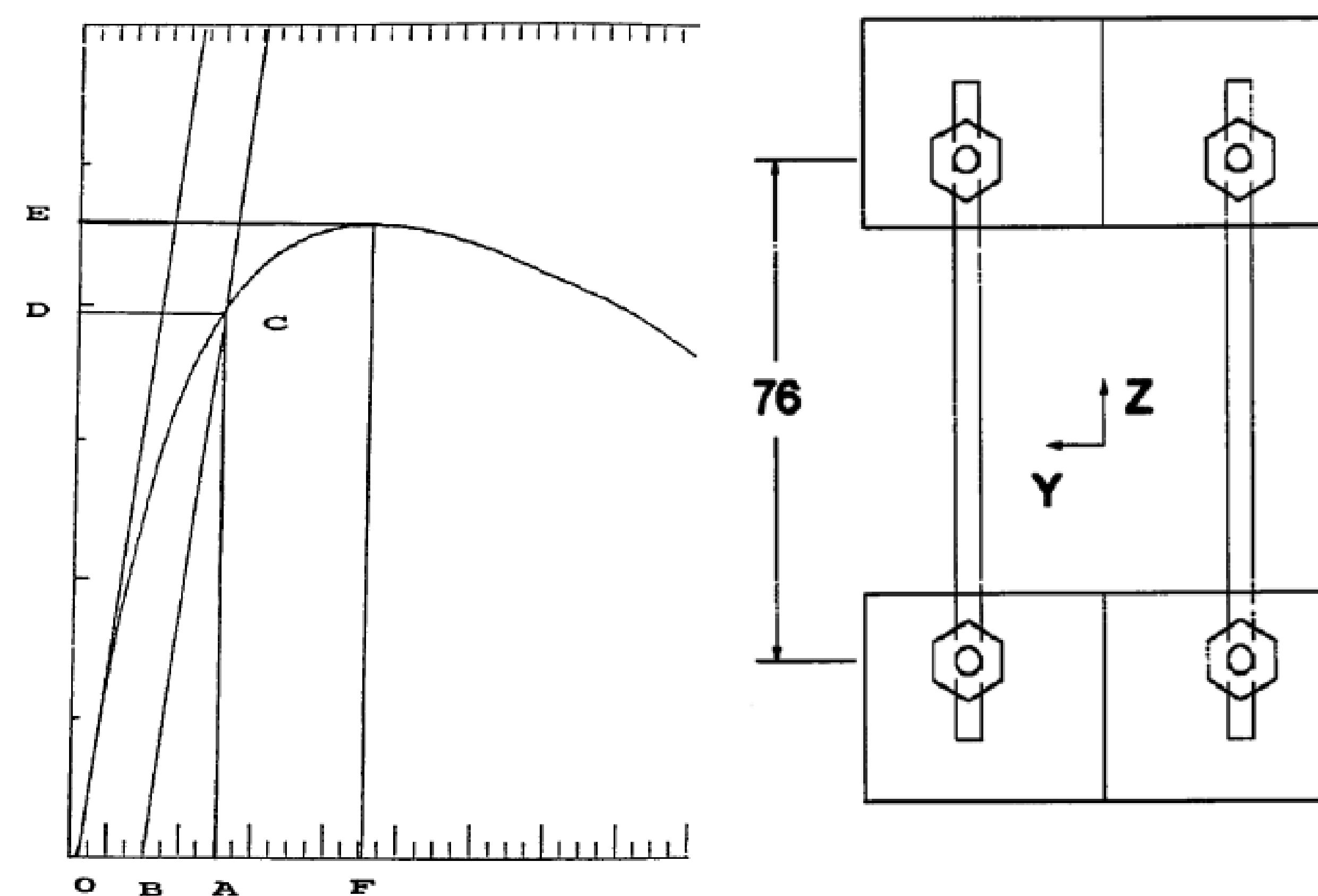


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METHODS

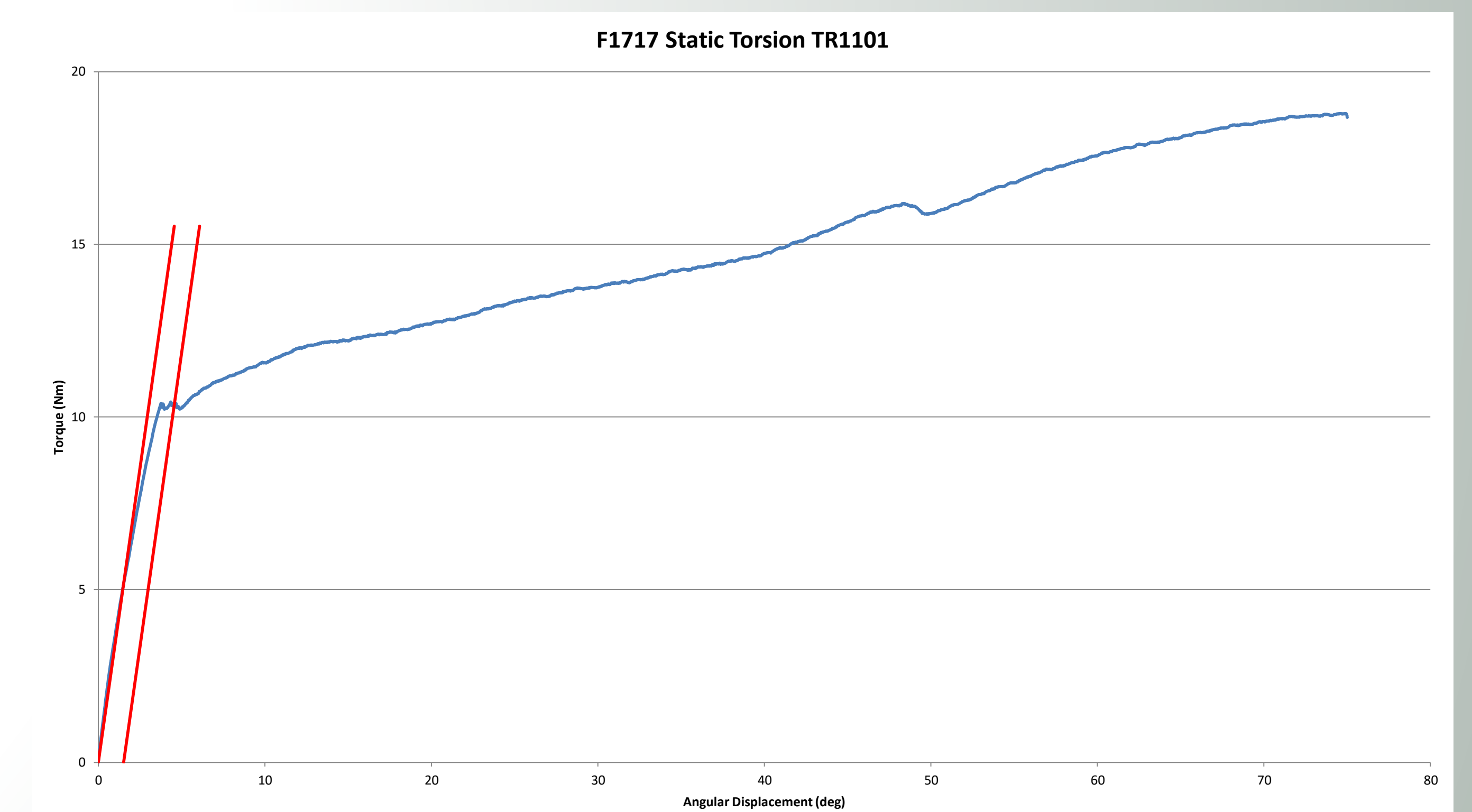
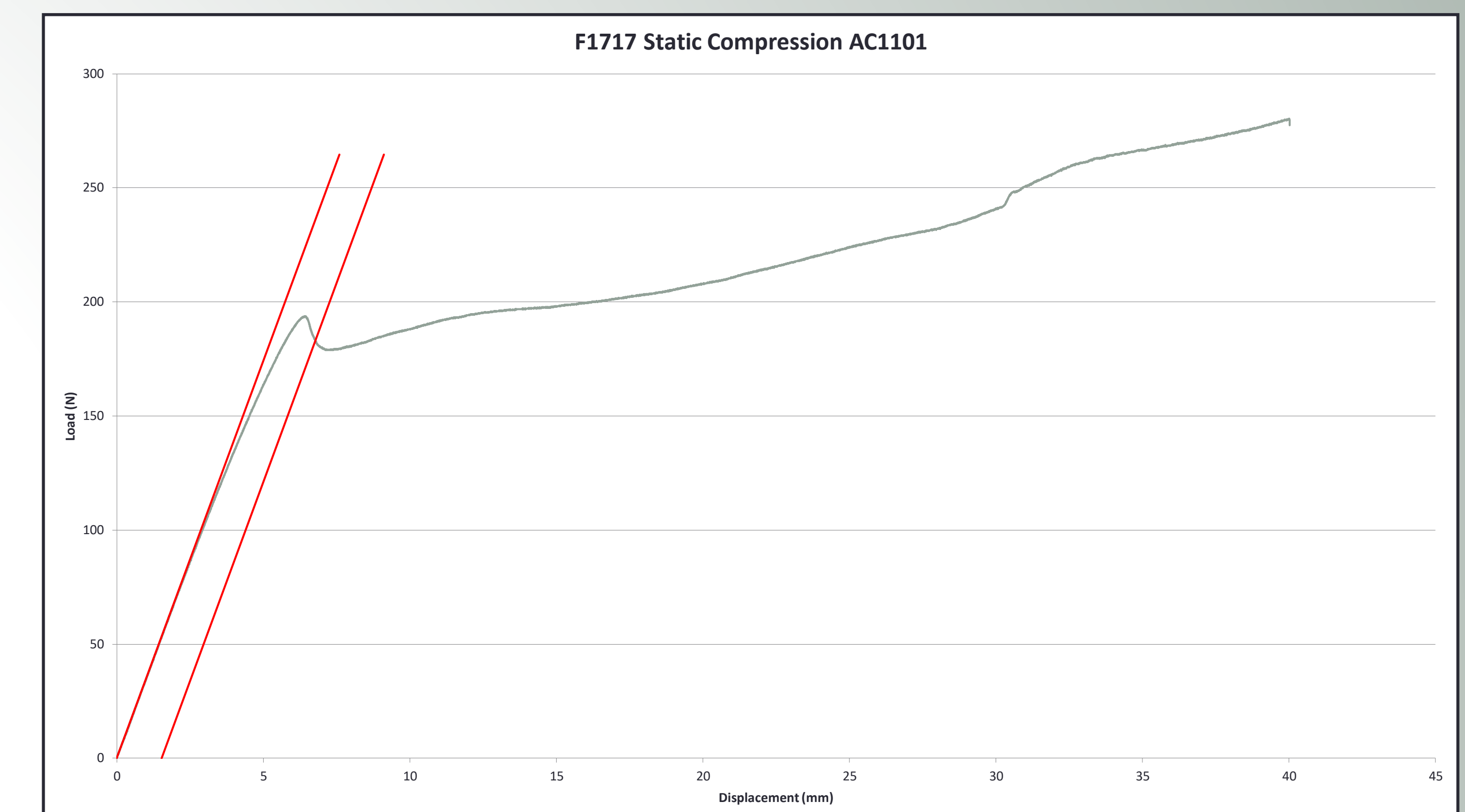
- Followed the ASTM standard for testing spinal implants
- Developed the standard operating procedures for testing these devices in the INSTRON materials testing machine
 - Preformed preliminary tests to develop the standard operating procedures
 - Followed the ASTM F1717 to develop methodology
 - The dimensions from the image bellow was used to design the preliminary tests
- Used the data collected from the INSTRON to determine values for compression and torsional testing
 - 2% offset yield, Elastic displacement, Compression bending peak load, Compression bending peak moment

- Images from the ASTM F1717 standard describing what values on the graph means what and the proper dimensions for the testing apparatus



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RESULTS



CONCLUSIONS AND DISCUSSION

- Compared to a previous study, the data was comparable and was overall considered to be acceptable by ASTM standards
- The final SOP developed was used to preform the actual testing to derive the needed data
- The ASTM standard was successfully used to develop these tests and acquire the appropriate data